part f f1 next

Formulas & Concepts to Take Note Of

1. **ϵ-Neighborhood of a Point p ($N_\epsilon(p)$):**
   The set of all points q whose Euclidean distance to p is less than or equal to ϵ, including p itself:

   MLSys_Worksheet2_Clas... + 3

   $$N_\epsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \epsilon\}$$

   MLSys_Worksheet2_Clas... + 1
   1. Helpful distance benchmark for $\epsilon = 1.5$:
      MLSys_Worksheet2_Clas... + 1
      1. Horizontal/Vertical adjacent (distance $= 1 \leq 1.5$): **Included.**
         MLSys_Worksheet2_Clas... + 1
      2. Diagonal neighbor ($\text{dist} = \quad 1^2 + 1^2 = \quad 2 \approx 1.414 \leq 1.5$): **Included.**
         MLSys_Worksheet2_Clas... + 1
      3. Distance ≥ 2: **Excluded.**
         MLSys_Worksheet2_Clas... + 1
2. **DBSCAN Point Classifications:**
   MLSys_Worksheet2_Clas... + 1
   1. **Core Point:** If its neighborhood contains at least MinPts points:

      MLSys_Worksheet2_Clas... + 1

      $$|N_\epsilon(p)| \geq \text{MinPts}$$

      MLSys_Worksheet2_Clas... + 1
   2. **Border Point:** Not a core point ($|N_\epsilon(p)| < \text{MinPts}$), but falls within the ϵ-neighborhood of at least one Core point.
      MLSys_Worksheet2_Clas... + 1
   3. **Noise Point:** Neither a core point nor within distance ϵ of any core point.
      MLSys_Worksheet2_Clas... + 1
3. **Cluster Formation (Density-Reachability):**

1. A cluster is formed by connecting core points that are within ϵ distance of each other (density-connected).
   MLSys_Worksheet2_Clas... +1
2. Any border point within distance ϵ of a core point is absorbed into that core point's cluster.
   MLSys_Worksheet2_Clas... +1

Given Parameters (Question F1)

- **9 Data Points:**
  $P_1 = (1,1), P_2 = (2,2), P_3 = (3,2), P_4 = (4,3), P_5 = (7,5), P_6 = (8,4), P_7 = (9,5), P_8 = (14,2), P_9 = (15,2).$
  MLSys_Worksheet2_Clas... +1
- $\epsilon = 1.5$
  MLSys_Worksheet2_Clas... +1
- $\text{MinPts} = 3$
  MLSys_Worksheet2_Clas... +1

Step-by-Step Solutions (Question F1)

(a) Complete the table for $\epsilon = 1.5, \text{MinPts} = 3$

Let us find $N_{1.5}(p)$ for every point:

- $P_1(1,1)$:
  - Distance to $P_2(2,2) = \quad (2-1)^2 + (2-1)^2 = \quad 2 \approx 1.414 \leq 1.5.$
    MLSys_Worksheet2_Clas... +1
  - Distance to $P_3(3,2) = \quad 2^2 + 1^2 = \quad 5 \approx 2.24 > 1.5.$
  - Neighborhood: $\{P_1, P_2\} \implies |N_\epsilon(P_1)| = 2.$
- $P_2(2,2)$:
  - Distance to $P_1(1,1) = \quad 2 \approx 1.414 \leq 1.5.$
    MLSys_Worksheet2_Clas... +1
  - Distance to $P_3(3,2) = \quad (3-2)^2 + (2-2)^2 = 1.0 \leq 1.5.$
  - Distance to $P_4(4,3) = \quad 2^2 + 1^2 = \quad 5 > 1.5.$
  - Neighborhood: $\{P_1, P_2, P_3\} \implies |N_\epsilon(P_2)| = 3$ (Core!).
- $P_3(3,2)$:
  - Distance to $P_2(2,2) = 1.0 \leq 1.5.$
  - Distance to $P_4(4,3) = \quad (4-3)^2 + (3-2)^2 = \quad 2 \approx 1.414 \leq 1.5.$
    MLSys_Worksheet2_Clas... +1

- Distance to $P_1 = \quad 5 > 1.5$.
- Neighborhood: $\{P_2, P_3, P_4\} \implies |N_\epsilon(P_3)| = 3$ (Core!).

- $P_4(4, 3)$:
  - Distance to $P_3(3, 2) = \quad 2 \approx 1.414 \leq 1.5$.
    MLSys_Worksheet2_Clas... +1
  - Nearest next point is $P_5(7, 5)$: $\text{dist} = \quad 3^2 + 2^2 = \quad 13 \approx 3.61 > 1.5$.
  - Neighborhood: $\{P_3, P_4\} \implies |N_\epsilon(P_4)| = 2$.
- $P_5(7, 5)$:
  - Distance to $P_6(8, 4) = \quad (8-7)^2 + (4-5)^2 = \quad 1 + 1 = \quad 2 \approx 1.414 \leq 1.5$.
    MLSys_Worksheet2_Clas... +1
  - Distance to $P_7(9, 5) = \quad (9-7)^2 + 0 = 2.0 > 1.5$.
    MLSys_Worksheet2_Clas... +1
  - Neighborhood: $\{P_5, P_6\} \implies |N_\epsilon(P_5)| = 2$.
- $P_6(8, 4)$:
  - Distance to $P_5(7, 5) = \quad 2 \approx 1.414 \leq 1.5$.
    MLSys_Worksheet2_Clas... +1
  - Distance to $P_7(9, 5) = \quad (9-8)^2 + (5-4)^2 = \quad 2 \approx 1.414 \leq 1.5$.
    MLSys_Worksheet2_Clas... +1
  - Neighborhood: $\{P_5, P_6, P_7\} \implies |N_\epsilon(P_6)| = 3$ (Core!).
- $P_7(9, 5)$:
  - Distance to $P_6(8, 4) = \quad 2 \approx 1.414 \leq 1.5$.
    MLSys_Worksheet2_Clas... +1
  - Distance to $P_5(7, 5) = 2.0 > 1.5$.
    MLSys_Worksheet2_Clas... +1
  - Neighborhood: $\{P_6, P_7\} \implies |N_\epsilon(P_7)| = 2$.
- $P_8(14, 2)$:
  - Distance to $P_9(15, 2) = \quad (15-14)^2 + 0 = 1.0 \leq 1.5$.
  - Neighborhood: $\{P_8, P_9\} \implies |N_\epsilon(P_8)| = 2$.
- $P_9(15, 2)$:
  - Distance to $P_8(14, 2) = 1.0 \leq 1.5$.
  - Neighborhood: $\{P_8, P_9\} \implies |N_\epsilon(P_9)| = 2$.

Determine Labels:

- **Core points** ($|N_\epsilon| \geq 3$): P_2, P_3, P_6.
  MLSys_Worksheet2_Clas... +1
- **Border points** ($|N_\epsilon| < 3$, but adjacent to a core point):

- P_1 is within 1.5 of core point P_2 $\implies$ **Border**.
  MLSys_Worksheet2_Clas... +1
- P_4 is within 1.5 of core point P_3 $\implies$ **Border**.
  MLSys_Worksheet2_Clas... +1
- P_5 is within 1.5 of core point P_6 $\implies$ **Border**.
  MLSys_Worksheet2_Clas... +1
- P_7 is within 1.5 of core point P_6 $\implies$ **Border**.
  MLSys_Worksheet2_Clas... +1

- **Noise points** ($|N_\epsilon| < 3$ and not adjacent to any core point):
  - P_8 and P_9 have size 2, but neither P_8 nor P_9 is a core point $\implies$ **Noise**.
    MLSys_Worksheet2_Clas... +1

Point p MLSys_Worksheet2_Clas... +1	$N_\epsilon(p)$ MLSys_Worksheet2_Clas... +1	$\lvert N_\epsilon(p)\rvert$ MLSys_Worksheet2_Clas... +1	Labe Nois MLS
$P_1(1,1)$ MLSys_Worksheet2_Clas... +1	$\{P_1, P_2\}$	2	Bord MLS
$P_2(2,2)$ MLSys_Worksheet2_Clas... +1	$\{P_1, P_2, P_3\}$	3	Core MLS
$P_3(3,2)$ MLSys_Worksheet2_Clas... +1	$\{P_2, P_3, P_4\}$	3	Core MLS
$P_4(4,3)$ MLSys_Worksheet2_Clas... +1	$\{P_3, P_4\}$	2	Bord MLS
$P_5(7,5)$ MLSys_Worksheet2_Clas... +1	$\{P_5, P_6\}$	2	Bord MLS
$P_6(8,4)$ MLSys_Worksheet2_Clas... +1	$\{P_5, P_6, P_7\}$	3	Core MLS
$P_7(9,5)$ MLSys_Worksheet2_Clas... +1	$\{P_6, P_7\}$	2	Bord MLS
$P_8(14,2)$ MLSys_Worksheet2_Clas... +1	$\{P_8, P_9\}$	2	Nois MLS
$P_9(15,2)$ MLSys_Worksheet2_Clas... +1	$\{P_8, P_9\}$	2	Nois MLS

(b) How many clusters are found, and which points are in each? Which points are noise?

- Core points P_2 and P_3 are within distance $1.0 \leq 1.5$, connecting into one density cluster along with their border points P_1 and P_4.
- Core point P_6 is far from P_2/P_3 ($\text{dist} > 1.5$), forming a second separate density cluster with its border points P_5 and P_7.
- P_8 and P_9 are isolated from all core points.

MLSys_Worksheet2_Clas... +1

Summary:

- Number of clusters found: 2 clusters

MLSys_Worksheet2_Clas... +1

- Cluster 1: $\{P_1, P_2, P_3, P_4\}$
- Cluster 2: $\{P_5, P_6, P_7\}$
- Noise points: $\{P_8, P_9\}$

MLSys_Worksheet2_Clas... +1

(c) Sensitivity Reruns

Case 1: Rerun with $\text{MinPts} = 2$ ($\epsilon = 1.5$)

- Since $|N_{1.5}(p)| \geq 2$ for **every single point**, every point now qualifies as a **Core point**.

MLSys_Worksheet2_Clas... +1

- P_8 and P_9 now have $|N_{1.5}| = 2 \geq 2$, so they become a **new 3rd cluster** $\{P_8, P_9\}$ instead of being marked as noise.

MLSys_Worksheet2_Clas... +1

- **Outcome: 3 Clusters** (Cluster 1: $\{P_1, P_2, P_3, P_4\}$, Cluster 2: $\{P_5, P_6, P_7\}$, Cluster 3: $\{P_8, P_9\}$) and **0 Noise**.

MLSys_Worksheet2_Clas... +1

Case 2: Rerun with $\epsilon = 4.5$ ($\text{MinPts} = 3$)

MLSys_Worksheet2_Clas... +4

- Expanding radius to $\epsilon = 4.5$ connects points that were previously far apart:

MLSys_Worksheet2_Clas... +1

  - Distance from $P_4(4, 3)$ to $P_5(7, 5)$ is $\quad 3^2 + 2^2 = \quad 13 \approx 3.61 \leq 4.5$. This bridges Cluster 1 and Cluster 2 together into a single large cluster!

MLSys_Worksheet2_Clas... +1

- Distance from $P_7(9,5)$ to $P_8(14,2)$ is $\quad 5^2+3^2= \quad 34 \approx 5.83 > 4.5$, so P_8 and P_9 remain separated from the main group.
  MLSys_Worksheet2_Clas... +1
- Between P_8 and P_9, their distance is 1.0, but neither has any other neighbor within 4.5, so $|N_{4.5}(P_8)| = 2$ and $|N_{4.5}(P_9)| = 2$. With $\mathrm{MinPts} = 3$, they still do not have 3 points and are not near any core point $\implies$ **Noise.**
  MLSys_Worksheet2_Clas... +1

Point MLSys_Worksheet2_Clas... +1	$\lvert N_\epsilon \rvert$ ($\epsilon = 1.5$) MLSys_Worksheet2_Clas... +1	Label ($\mathrm{MinPts} = 2$) MLSys_Worksheet2_Clas... +1	$N_\epsilon($ MLS
$P_1(1,1)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_2(2,2)$ MLSys_Worksheet2_Clas... +1	3 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_3(3,2)$ MLSys_Worksheet2_Clas... +1	3 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_4(4,3)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_5(7,5)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_3,$ MLS
$P_6(8,4)$ MLSys_Worksheet2_Clas... +1	3 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_4,$ MLS
$P_7(9,5)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_5,$ MLS
$P_8(14,2)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_8,$ MLS
$P_9(15,2)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_8,$ MLS

- **How clusters change:**
  - For $\mathrm{MinPts} = 2$: Density threshold drops, turning P_8, P_9 from noise into a **3rd cluster**.
    MLSys_Worksheet2_Clas... +1
  - For $\epsilon = 4.5$: Radius expands enough to merge Clusters 1 and 2 into **one single giant cluster** ($\{P_1, \ldots, P_7\}$), while P_8, P_9 remain **noise**.
    MLSys_Worksheet2_Clas... +1

can you explain f1 a bit easier pls

DBSCAN groups points based on **how crowded** an area is.

MLSys_Worksheet2_Clas... + 1

Here are the basic terms to know before solving:

- $\epsilon = 1.5$ **(The Reach / Circle):** Imagine drawing a circle of radius 1.5 around each point. Any other point falling inside that circle counts as a neighbor.
  MLSys_Worksheet2_Clas... + 3
- $\text{MinPts} = 3$ **(The Crowd Threshold):** A point needs **at least 3 points** inside its circle (counting itself!) to be considered a dense center.
  MLSys_Worksheet2_Clas... + 1

The Three Roles in DBSCAN

1. **Core Point:** A popular spot. Its circle contains ≥ 3 **points** (itself $+2$ or more friends).
   MLSys_Worksheet2_Clas... + 3
2. **Border Point:** A hanger-on. It has fewer than 3 points in its own circle, but it is close enough to touch a **Core point**. It gets invited to the cluster because of its Core friend.
   MLSys_Worksheet2_Clas... + 4
3. **Noise:** An outcast. It has fewer than 3 points in its circle, and **none** of its neighbors are Core points either. It belongs to no cluster.
   MLSys_Worksheet2_Clas... + 4

Quick Trick to Measure Distances

Because $\epsilon = 1.5$:

MLSys_Worksheet2_Clas... + 1

- Points directly above, below, or next to each other by 1 step (like $(14, 2)$ to $(15, 2)$) have a distance of $1.0 \implies$ **Within circle** (≤ 1.5).
  MLSys_Worksheet2_Clas... + 1
- Points diagonally 1 step away (like $(1, 1)$ to $(2, 2)$) have a distance of $1^2 + 1^2 = 2 \approx \implies$ **Within circle** (≤ 1.5).
  MLSys_Worksheet2_Clas... + 1
- Any gap of 2 steps or more is too far ($\geq 2 > 1.5$).
  MLSys_Worksheet2_Clas... + 1

Step-by-Step for Each Point

1. **Look at the first group:** $P_1(1,1), P_2(2,2), P_3(3,2), P_4(4,3)$
   MLSys_Worksheet2_Clas... + 1
   1. $P_1(1,1)$: Only touches itself and diagonal neighbor P_2.
      MLSys_Worksheet2_Clas... + 1
      1. Count $= 2$. Since $2 < 3$, it is not Core. But because P_2 is a Core point, P_1 becomes a **Border point**.
         MLSys_Worksheet2_Clas... + 4
   2. $P_2(2,2)$: Touches P_1, itself, and P_3.
      MLSys_Worksheet2_Clas... + 1
      1. Count $= 3$. It hits $\text{MinPts} = 3 \implies$ **Core point**!
         MLSys_Worksheet2_Clas... + 3
   3. $P_3(3,2)$: Touches P_2, itself, and P_4.
      MLSys_Worksheet2_Clas... + 1
      1. Count $= 3$. It hits $\text{MinPts} = 3 \implies$ **Core point**!
         MLSys_Worksheet2_Clas... + 3
   4. $P_4(4,3)$: Touches P_3 and itself.
      MLSys_Worksheet2_Clas... + 1
      1. Count $= 2$. Since $2 < 3$, it is not Core. But because neighbor P_3 is a Core point, P_4 becomes a **Border point**.
         MLSys_Worksheet2_Clas... + 4

   👉 **Cluster 1** forms here: Core points P_2 and P_3 connect with each other, pulling in border points P_1 and P_4.
   MLSys_Worksheet2_Clas... + 1
   Members: $\{P_1, P_2, P_3, P_4\}$.
   MLSys_Worksheet2_Clas... + 1

1. **Look at the second group:** $P_5(7,5), P_6(8,4), P_7(9,5)$
   MLSys_Worksheet2_Clas... + 1
   1. $P_6(8,4)$: Sits in the middle. It touches diagonal neighbors P_5 and P_7, plus itself.
      MLSys_Worksheet2_Clas... + 3
      1. Count $= 3 \implies$ **Core point**!
         MLSys_Worksheet2_Clas... + 3
   2. $P_5(7,5)$: Only touches P_6 and itself.
      MLSys_Worksheet2_Clas... + 1
      1. Count $= 2$. Touches Core point $P_6 \implies$ **Border point**.
         MLSys_Worksheet2_Clas... + 3
   3. $P_7(9,5)$: Only touches P_6 and itself.
      MLSys_Worksheet2_Clas... + 1
      1. Count $= 2$. Touches Core point $P_6 \implies$ **Border point**.
         MLSys_Worksheet2_Clas... + 3

   👉 **Cluster 2** forms here: Core point P_6 pulls in its borders P_5 and P_7.
   MLSys_Worksheet2_Clas... + 1
   Members: $\{P_5, P_6, P_7\}$.
   MLSys_Worksheet2_Clas... + 1

1. **Look at the far points: $P_8(14, 2)$ and $P_9(15, 2)$**
   MLSys_Worksheet2_Clas... +1
   1. They are right next to each other (distance $= 1.0$), but way out at $x = 14$ and 15, far from everyone else.
      MLSys_Worksheet2_Clas... +1
   2. P_8 touches P_9 and itself (Count $= 2$).
      MLSys_Worksheet2_Clas... +1
   3. P_9 touches P_8 and itself (Count $= 2$).
      MLSys_Worksheet2_Clas... +1
   4. Neither reaches the threshold of 3. Since **neither is a Core point**, neither can save the other!
      MLSys_Worksheet2_Clas... +3

   👉 Both are abandoned as **Noise**.
   MLSys_Worksheet2_Clas... +1

What happens in the "What-if" reruns (Letter c)?

- **What if we lower MinPts from 3 to 2?**
  MLSys_Worksheet2_Clas... +1
  - Now you only need 2 points to be a Core point!
    MLSys_Worksheet2_Clas... +1
  - Every single point in the dataset has at least 2 points in its circle, so **everyone becomes Core.**
    MLSys_Worksheet2_Clas... +1
  - P_8 and P_9 are no longer noise; they form their own tiny **Cluster 3.**
    MLSys_Worksheet2_Clas... +1
- **What if we enlarge the circle to $\epsilon = 4.5$?**
  MLSys_Worksheet2_Clas... +1
  - The reach is now large enough to jump the empty gap between $P_4(4, 3)$ and $P_5(7, 5)$ (the distance across that gap is about 3.6).
    MLSys_Worksheet2_Clas... +1
  - Cluster 1 and Cluster 2 merge into **one single giant cluster**.
    MLSys_Worksheet2_Clas... +1
  - However, P_8 and P_9 are still too far away (distance > 5.8), so with $\mathrm{MinPts} = 3$, they stay **Noise.**
    MLSys_Worksheet2_Clas... +1

show calculations too

Here is the complete arithmetic calculation for every single pairwise distance in **F1.**

MLSys_Worksheet2_Clas... +1

Distance Formula

For any two 2D points $P_a(x_a, y_a)$ and $P_b(x_b, y_b)$, the Euclidean distance is:

MLSys_Worksheet2_Clas... +1

$$\text{dist}(P_a, P_b) = \quad (x_b - x_a)^2 + (y_b - y_a)^2$$

MLSys_Worksheet2_Clas... +1

To check if P_b is inside the circle of radius ϵ, test if $\text{dist}(P_a, P_b) \leq \epsilon$, or equivalently:

MLSys_Worksheet2_Clas... +1

$$\text{dist}^2 \leq \epsilon^2$$

- For $\epsilon = 1.5$: $\epsilon^2 = 1.5^2 =$ (any point with squared distance ≤ 2.25 is a neighbor).
  MLSys_Worksheet2_Clas... +1
- For $\epsilon = 4.5$: $\epsilon^2 = 4.5^2 =$ (any point with squared distance ≤ 20.25 is a neighbor).
  MLSys_Worksheet2_Clas... +1

1. Calculations for Part (a) $[\epsilon = 1.5, \text{MinPts} = 3]$

$P_1(1, 1)$

- $\text{dist}^2(P_1, P_1) = 0 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_1, P_2) = (2-1)^2 + (2-1)^2 = 1^2 + 1^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle ($\text{dist} = \quad 2 \approx 1.41$)
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_1, P_3) = (3-1)^2 + (2-1)^2 = 2^2 + 1^2 = 4 + 1 = 5 > 2.25 \implies \text{Out}$
- (All remaining points $P_4 \ldots P_9$ have $x \geq 4$, so $\Delta x^2 \geq 3^2 = 9 > 2.25 \implies \text{Out}$)
- **Neighborhood:** $N_{1.5}(P_1) = \{P_1, P_2\}$

MLSys_Worksheet2_Clas... +1

- Count: $|N_{1.5}(P_1)| =$

MLSys_Worksheet2_Clas... +1

$P_2(2, 2)$

- $\text{dist}^2(P_2, P_1) = (1-2)^2 + (1-2)^2 = (-1)^2 + (-1)^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle

MLSys_Worksheet2_Clas... +1

- $\text{dist}^2(P_2, P_2) = 0 \leq 2.25 \implies$ is in circle

MLSys_Worksheet2_Clas... +1

- $\text{dist}^2(P_2, P_3) = (3-2)^2 + (2-2)^2 = 1^2 + 0^2 = 1 + 0 = 1 \leq 2.25 \implies$ is in circle ($\text{dist} = \sqrt{1} = 1.0$)

MLSys_Worksheet2_Clas... +1

- $\text{dist}^2(P_2, P_4) = (4-2)^2 + (3-2)^2 = 2^2 + 1^2 = 4 + 1 = 5 > 2.25 \implies$ Out
- Neighborhood: $N_{1.5}(P_2) = \{P_1, P_2, P_3\}$

MLSys_Worksheet2_Clas... +1

- Count: $|N_{1.5}(P_2)| =$

MLSys_Worksheet2_Clas... +1

$P_3(3, 2)$

- $\text{dist}^2(P_3, P_1) = (1-3)^2 + (1-2)^2 = (-2)^2 + (-1)^2 = 4 + 1 = 5 > 2.25 \implies$ Out
- $\text{dist}^2(P_3, P_2) = (2-3)^2 + (2-2)^2 = (-1)^2 + 0 = 1 \leq 2.25 \implies$ is in circle

MLSys_Worksheet2_Clas... +1

- $\text{dist}^2(P_3, P_3) = 0 \leq 2.25 \implies$ is in circle

MLSys_Worksheet2_Clas... +1

- $\text{dist}^2(P_3, P_4) = (4-3)^2 + (3-2)^2 = 1^2 + 1^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle ($\text{dist} = \sqrt{2} \approx 1.41$)

MLSys_Worksheet2_Clas... +1

- $\text{dist}^2(P_3, P_5) = (7-3)^2 + (5-2)^2 = 4^2 + 3^2 = 16 + 9 = 25 > 2.25 \implies$ Out
- Neighborhood: $N_{1.5}(P_3) = \{P_2, P_3, P_4\}$

MLSys_Worksheet2_Clas... +1

- Count: $|N_{1.5}(P_3)| =$

MLSys_Worksheet2_Clas... +1

$P_4(4, 3)$

- $\text{dist}^2(P_4, P_2) = (2-4)^2 + (2-3)^2 = (-2)^2 + (-1)^2 = 4 + 1 = 5 > 2.25 \implies \text{Out}$
- $\text{dist}^2(P_4, P_3) = (3-4)^2 + (2-3)^2 = (-1)^2 + (-1)^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_4, P_4) = 0 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_4, P_5) = (7-4)^2 + (5-3)^2 = 3^2 + 2^2 = 9 + 4 = 13 > 2.25 \implies \text{Out}$
- Neighborhood: $N_{1.5}(P_4) = \{P_3, P_4\}$
  MLSys_Worksheet2_Clas... +1
- Count: $|N_{1.5}(P_4)| =$
  MLSys_Worksheet2_Clas... +1

$P_5(7, 5)$

- $\text{dist}^2(P_5, P_4) = 13 > 2.25 \implies \text{Out}$
- $\text{dist}^2(P_5, P_5) = 0 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_5, P_6) = (8-7)^2 + (4-5)^2 = 1^2 + (-1)^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle ($\text{dist} = \sqrt{2} \approx 1.41$)
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_5, P_7) = (9-7)^2 + (5-5)^2 = 2^2 + 0^2 = 4 + 0 = 4 > 2.25 \implies \text{Out}$ ($\text{dist} = 2.0 > 1.5$)
  MLSys_Worksheet2_Clas... +1
- Neighborhood: $N_{1.5}(P_5) = \{P_5, P_6\}$
  MLSys_Worksheet2_Clas... +1
- Count: $|N_{1.5}(P_5)| =$
  MLSys_Worksheet2_Clas... +1

$P_6(8, 4)$

- $\text{dist}^2(P_6, P_5) = (7-8)^2 + (5-4)^2 = (-1)^2 + 1^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_6, P_6) = 0 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_6, P_7) = (9-8)^2 + (5-4)^2 = 1^2 + 1^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle ($\text{dist} = \sqrt{2} \approx 1.41$)
  MLSys_Worksheet2_Clas... +1

- Neighborhood: $N_{1.5}(P_6) = \{P_5, P_6, P_7\}$
  MLSys_Worksheet2_Clas... +1
- Count: $|N_{1.5}(P_6)| =$
  MLSys_Worksheet2_Clas... +1

$P_7(9, 5)$

- $\text{dist}^2(P_7, P_5) = (7-9)^2 + (5-5)^2 = (-2)^2 + 0 = 4 > 2.25 \implies \text{Out}$
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_7, P_6) = (8-9)^2 + (4-5)^2 = (-1)^2 + (-1)^2 = 1 + 1 = 2 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_7, P_7) = 0 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_7, P_8) = (14-9)^2 + (2-5)^2 = 5^2 + (-3)^2 = 25 + 9 = 34 > 2.25 \implies \text{Out}$
- Neighborhood: $N_{1.5}(P_7) = \{P_6, P_7\}$
  MLSys_Worksheet2_Clas... +1
- Count: $|N_{1.5}(P_7)| =$
  MLSys_Worksheet2_Clas... +1

$P_8(14, 2)$

- $\text{dist}^2(P_8, P_8) = 0 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_8, P_9) = (15-14)^2 + (2-2)^2 = 1^2 + 0^2 = 1 \leq 2.25 \implies$ is in circle ($\text{dist} = 1.0$)
  MLSys_Worksheet2_Clas... +1
- Neighborhood: $N_{1.5}(P_8) = \{P_8, P_9\}$
  MLSys_Worksheet2_Clas... +1
- Count: $|N_{1.5}(P_8)| =$
  MLSys_Worksheet2_Clas... +1

$P_9(15, 2)$

- $\text{dist}^2(P_9, P_8) = (14-15)^2 + (2-2)^2 = (-1)^2 + 0 = 1 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- $\text{dist}^2(P_9, P_9) = 0 \leq 2.25 \implies$ is in circle
  MLSys_Worksheet2_Clas... +1
- Neighborhood: $N_{1.5}(P_9) = \{P_8, P_9\}$
  MLSys_Worksheet2_Clas... +1

- **Count:** $|N_{1.5}(P_9)| =$

MLSys_Worksheet2_Clas... +1

Classification Checks ($\mathrm{MinPts} = 3$)

1. Points with count ≥ 3: $P_2, P_3, P_6 \implies$ **Core**.

MLSys_Worksheet2_Clas... +1

2. For points with count $= 2$:
    1. P_1 contains P_2 (which is Core) $\implies$

    MLSys_Worksheet2_Clas... +1
    2. P_4 contains P_3 (which is Core) $\implies$

    MLSys_Worksheet2_Clas... +1
    3. P_5 contains P_6 (which is Core) $\implies$

    MLSys_Worksheet2_Clas... +1
    4. P_7 contains P_6 (which is Core) $\implies$

    MLSys_Worksheet2_Clas... +1
    5. P_8 contains only $\{P_8, P_9\}$ (neither is Core) $\implies$

    MLSys_Worksheet2_Clas... +1
    6. P_9 contains only $\{P_8, P_9\}$ (neither is Core) $\implies$

    MLSys_Worksheet2_Clas... +1

2. Calculations for Part (c) [Enlarged Radius $\epsilon = 4.5 \implies \epsilon^2 = 20.25$]

MLSys_Worksheet2_Clas... +4

Let us compute the critical bridging distances between the clusters:

MLSys_Worksheet2_Clas... +1

- **Distance between Group 1 and Group 2:**

$$\mathrm{dist}^2(P_4, P_5) = (7-4)^2 + (5-3)^2 = 3^2 + 2^2 = 9 + 4 = \quad \leq 20$$

MLSys_Worksheet2_Clas...

$$\mathrm{dist}^2(P_3, P_5) = (7-3)^2 + (5-2)^2 = 4^2 + 3^2 = 16 + 9 = \quad > 2$$

$$\mathrm{dist}^2(P_4, P_6) = (8-4)^2 + (4-3)^2 = 4^2 + 1^2 = 16 + 1 = \quad \leq 2$$

(Because P_4 is within 4.5 of both P_5 and P_6, the two clusters are linked together!)

MLSys_Worksheet2_Clas... +1

- **Distance between Group 1 points:**

$$\text{dist}^2(P_1, P_4) = (4-1)^2 + (3-1)^2 = 3^2 + 2^2 = 9 + 4 = \quad \leq 20$$

(All points in $\{P_1, P_2, P_3, P_4\}$ are within 4.5 of each other, giving every one of them a count of at least 4).

MLSys_Worksheet2_Clas... +1

- **Distance between Group 2 and the far points P_8, P_9:**

$$\text{dist}^2(P_7, P_8) = (14-9)^2 + (2-5)^2 = 5^2 + (-3)^2 = 25 + 9 =$$

$$\text{dist}(P_7, P_8) = \quad 34 \approx \qquad > 4.5 \implies \text{Out}$$

$$\text{dist}^2(P_6, P_8) = (14-8)^2 + (2-4)^2 = 6^2 + (-2)^2 = 36 + 4 =$$

- **Result for P_8 and P_9 with $\epsilon = 4.5$:**

$$\text{dist}^2(P_8, P_9) = 1 \leq 20.25 \implies \text{Only } \{P_8, P_9\} \text{ are within reach o}$$

MLSys_Worksheet2_Clas...

$$|N_{4.5}(P_8)| = 2 \quad \text{and} \quad |N_{4.5}(P_9)| = 2$$

Since $\text{MinPts} = 3$ and $2 < 3$, with no other core points nearby, P_8 and P_9 remain **Noise.**

MLSys_Worksheet2_Clas... +1

letter c next

Here are the detailed steps, complete arithmetic calculations, and answers for **F1 (c)**.

MLSys_Worksheet2_Clas... +1

Understanding the Two Reruns in (c)

In DBSCAN, tuning the hyperparameters changes how clusters form:

MLSys_Worksheet2_Clas... +1

1. **Changing MinPts from $3 \to 2$ (keeping $\epsilon = 1.5$):**
    1. The distance circle size (ϵ) remains the same.
    MLSys_Worksheet2_Clas... +1
    2. The threshold to become a **Core point** drops from needing 3 points down to needing only 2 **points**.
    MLSys_Worksheet2_Clas... +1
2. **Changing ϵ from $1.5 \to 4.5$ (keeping $\mathrm{MinPts} = 3$):**
    1. The threshold for a Core point remains 3 **points**.
    MLSys_Worksheet2_Clas... +1
    2. The radius of each point's neighborhood circle expands significantly, from $\epsilon = 1.5$ ($\epsilon^2 = 2.25$) to $\epsilon = 4.5$ ($\epsilon^2 = 20.25$).
    MLSys_Worksheet2_Clas... +1

Step-by-Step Calculations

1. Case 1: $\mathrm{MinPts} = 2$ with $\epsilon = 1.5$

From part (a), the neighborhood counts $|N_{1.5}(p)|$ were already calculated:

MLSys_Worksheet2_Clas... +1

- $|N_{1.5}(P_1)| = 2$
MLSys_Worksheet2_Clas...
- $|N_{1.5}(P_2)| = 3$
MLSys_Worksheet2_Clas...
- $|N_{1.5}(P_3)| = 3$
MLSys_Worksheet2_Clas...
- $|N_{1.5}(P_4)| = 2$
MLSys_Worksheet2_Clas...

- $|N_{1.5}(P_5)| = 2$
  MLSys_Worksheet2_Clas...
- $|N_{1.5}(P_6)| = 3$
  MLSys_Worksheet2_Clas...
- $|N_{1.5}(P_7)| = 2$
  MLSys_Worksheet2_Clas...
- $|N_{1.5}(P_8)| = 2$
  MLSys_Worksheet2_Clas...
- $|N_{1.5}(P_9)| = 2$
  MLSys_Worksheet2_Clas...

Label Evaluation ($\mathrm{MinPts} = 2$):

- Any point with $|N_{1.5}(p)| \geq 2$ is now a **Core point.**
  MLSys_Worksheet2_Clas... +1
- Since **every point** has a count of at least 2, **all 9 points become Core points**!
  MLSys_Worksheet2_Clas...

2. Case 2: $\epsilon = 4.5$ ($\epsilon^2 = 20.25$) with $\mathrm{MinPts} = 3$

Compute the squared Euclidean distance $\mathrm{dist}^2 = (x_b - x_a)^2 + (y_b - y_a)^2$ from each point to find all points where $\mathrm{dist}^2 \leq 20.25$:

MLSys_Worksheet2_Clas...

- For $P_1(1, 1)$:
  - to $P_2(2, 2)$: $(1)^2 + (1)^2 = 2 \leq 20.25 \implies$ **Included**
  - to $P_3(3, 2)$: $(2)^2 + (1)^2 = 5 \leq 20.25 \implies$ **Included**
  - to $P_4(4, 3)$: $(3)^2 + (2)^2 = 9 + 4 = 13 \leq 20.25 \implies$ **Included**
    MLSys_Worksheet2_Clas...
  - to $P_5(7, 5)$: $(6)^2 + (4)^2 = 36 + 16 = 52 > 20.25 \implies$ Out
  - $N_{4.5}(P_1) = \{P_1, P_2, P_3, P_4\} \implies |N_{4.5}(P_1)| =$
    MLSys_Worksheet2_Clas...
- For $P_2(2, 2)$:
  - to P_1: $2 \leq 20.25 \implies$ **Included**
  - to P_3: $(1)^2 + 0 = 1 \leq 20.25 \implies$ **Included**
  - to P_4: $(2)^2 + (1)^2 = 5 \leq 20.25 \implies$ **Included**
  - to P_5: $(5)^2 + (3)^2 = 25 + 9 = 34 > 20.25 \implies$ Out
  - $N_{4.5}(P_2) = \{P_1, P_2, P_3, P_4\} \implies |N_{4.5}(P_2)| =$
    MLSys_Worksheet2_Clas...
- For $P_3(3, 2)$:

    - to P_1: $5 \leq 20.25 \implies$ Included
    - to P_2: $1 \leq 20.25 \implies$ Included
    - to P_4: $(1)^2 + (1)^2 = 2 \leq 20.25 \implies$ Included
    - to $P_5(7,5)$: $(4)^2 + (3)^2 = 16 + 9 = 25 > 20.25 \implies$ Out
    - $N_{4.5}(P_3) = \{P_1, P_2, P_3, P_4\} \implies |N_{4.5}(P_3)| =$
    MLSys_Worksheet2_Clas...
- For $P_4(4,3)$ (The Bridge Point):
    - to P_1: $13 \leq 20.25 \implies$ Included
    MLSys_Worksheet2_Clas...
    - to P_2: $5 \leq 20.25 \implies$ Included
    - to P_3: $2 \leq 20.25 \implies$ Included
    - to $P_5(7,5)$: $(3)^2 + (2)^2 = 9 + 4 = 13 \leq 20.25 \implies$ Included
    MLSys_Worksheet2_Clas...
    - to $P_6(8,4)$: $(4)^2 + (1)^2 = 16 + 1 = 17 \leq 20.25 \implies$ Included
    - to $P_7(9,5)$: $(5)^2 + (2)^2 = 25 + 4 = 29 > 20.25 \implies$ Out
    - $N_{4.5}(P_4) = \{P_1, P_2, P_3, P_4, P_5, P_6\}$ *(Note: Teacher's key lists 5 points* $\{P_1, P_2, P_3, P_4, P_5\}$ *by checking only* P_5*)* $\implies |N_{4.5}(P_4)| =$
    MLSys_Worksheet2_Clas... +1
- For $P_5(7,5)$:
    - to $P_3(3,2)$: $(4)^2 + (3)^2 = 25 > 20.25 \implies$ Out
    MLSys_Worksheet2_Clas...
    - to $P_4(4,3)$: $13 \leq 20.25 \implies$ Included
    MLSys_Worksheet2_Clas...
    - to $P_6(8,4)$: $(1)^2 + (1)^2 = 2 \leq 20.25 \implies$ Included
    - to $P_7(9,5)$: $(2)^2 + 0 = 4 \leq 20.25 \implies$ Included
    - $N_{4.5}(P_5) = \{P_3, P_4, P_5, P_6, P_7\}$ *(or* $\{P_4, P_5, P_6, P_7\}$*)* $\implies |N_{4.5}(P_5)| =$ (or **4**)
    MLSys_Worksheet2_Clas...
- For $P_6(8,4)$:
    - to $P_4(4,3)$: $(4)^2 + (1)^2 = 17 \leq 20.25 \implies$ Included
    MLSys_Worksheet2_Clas...
    - to $P_5(7,5)$: $2 \leq 20.25 \implies$ Included
    - to $P_7(9,5)$: $(1)^2 + (1)^2 = 2 \leq 20.25 \implies$ Included
    - to $P_8(14,2)$: $(6)^2 + (2)^2 = 36 + 4 = 40 > 20.25 \implies$ Out
    - $N_{4.5}(P_6) = \{P_4, P_5, P_6, P_7\} \implies |N_{4.5}(P_6)| =$
    MLSys_Worksheet2_Clas...
- For $P_7(9,5)$:

- to $P_5(7,5)$: $4 \leq 20.25 \implies$ Included
  MLSys_Worksheet2_Clas...
- to $P_6(8,4)$: $2 \leq 20.25 \implies$ Included
  MLSys_Worksheet2_Clas...
- to $P_8(14,2)$: $(5)^2 + (3)^2 = 25 + 9 = 34 > 20.25 \implies$ Out
- $N_{4.5}(P_7) = \{P_5, P_6, P_7\} \implies |N_{4.5}(P_7)| =$
  MLSys_Worksheet2_Clas...

- For $P_8(14,2)$ & $P_9(15,2)$:
  - to each other: $(1)^2 + 0 = 1 \leq 20.25 \implies$ Included
  - to $P_7(9,5)$: $34 > 20.25 \implies$ Out
  - $N_{4.5}(P_8) = \{P_8, P_9\} \implies |N_{4.5}(P_8)| =$
    MLSys_Worksheet2_Clas...
  - $N_{4.5}(P_9) = \{P_8, P_9\} \implies |N_{4.5}(P_9)| =$
    MLSys_Worksheet2_Clas...

Label Evaluation ($\epsilon = 4.5, \mathrm{MinPts} = 3$):

- P_1 through P_7 each have counts $\geq 3 \implies$ **Core points.**
  MLSys_Worksheet2_Clas...
- P_8 and P_9 have counts $= 2 < 3$, and neither touches any core point (> 4.5 away from $P_1 \ldots P_7$) $\implies$ **Noise.**
  MLSys_Worksheet2_Clas...

Completed Table for (c)

Point	$\lvert N_\epsilon(p)\rvert$ ($\epsilon = 1.5$)	Label ($\mathrm{MinPts} = 2$)	$N_\epsilon($
MLSys_Worksheet2_Clas... +1	MLSys_Worksheet2_Clas... +1	MLSys_Worksheet2_Clas... +1	MLS
$P_1(1,1)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_2(2,2)$ MLSys_Worksheet2_Clas... +1	3 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_3(3,2)$ MLSys_Worksheet2_Clas... +1	3 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_4(4,3)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_1,$ MLS
$P_5(7,5)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	**Core** MLSys_Worksheet2_Clas...	$\{P_3,$ MLS

Point	$\lvert N_\epsilon(p)\rvert$ ($\epsilon = 1.5$)	Label ($\mathrm{MinPts} = 2$)	$N_\epsilon($
$P_6(8,4)$ MLSys_Worksheet2_Clas... +1	3 MLSys_Worksheet2_Clas...	Core MLSys_Worksheet2_Clas...	$\{P_4$ MLS
$P_7(9,5)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	Core MLSys_Worksheet2_Clas...	$\{P_5$ MLS
$P_8(14,2)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	Core MLSys_Worksheet2_Clas...	$\{P_8$ MLS
$P_9(15,2)$ MLSys_Worksheet2_Clas... +1	2 MLSys_Worksheet2_Clas...	Core MLSys_Worksheet2_Clas...	$\{P_8$ MLS

Written Answer: How Clusters and Noise Change

MLSys_Worksheet2_Clas... +4

- **When $\mathrm{MinPts} = 2$ ($\epsilon = 1.5$):**
  - Lowering the density threshold turns every single point into a Core point.
    MLSys_Worksheet2_Clas...
  - Points P_8 and P_9 change from noise into a **new 3rd cluster** $\{P_8, P_9\}$.
    MLSys_Worksheet2_Clas...
  - Total clusters increase from **2 to 3**, and noise decreases to **0**.
    MLSys_Worksheet2_Clas...
- **When $\epsilon = 4.5$ ($\mathrm{MinPts} = 3$):**
  - Expanding the search radius allows point P_4 to reach P_5 (distance $\approx 3.61 \leq 4.5$). This merges Cluster 1 and Cluster 2 into **one single large cluster** containing $\{P_1, P_2, P_3, P_4, P_5, P_6, P_7\}$.
    MLSys_Worksheet2_Clas... +1
  - Points P_8 and P_9 are still too far away (distance $> 5.8 > 4.5$), so they **remain Noise**.
    MLSys_Worksheet2_Clas...

why is e squared

Squaring ϵ is an algebraic shortcut that avoids computing square roots.

The Euclidean distance between two points (x_1, y_1) and (x_2, y_2) is:

$$\text{dist} = \quad (x_2 - x_1)^2 + (y_2 - y_1)^2$$

To check if a point is within distance ϵ, the condition is:

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 \leq \epsilon$$

Squaring both sides of the inequality removes the radical:

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 \leq \epsilon^2$$

Both comparisons produce the identical result because distances and radii are non-negative:

- For $\epsilon = 1.5$: $(1.5)^2 = 2.25$, so $(x_2 - x_1)^2 + (y_2 - y_1)^2 \leq 2.25$.
- For $\epsilon = 4.5$: $(4.5)^2 = 20.25$, so $(x_2 - x_1)^2 + (y_2 - y_1)^2 \leq 20.25$.

Using ϵ^2 allows you to check whether points are neighbors using basic addition and multiplication without calculating square roots for every pair of points.

f2 next

Formulas & Concepts to Take Note Of

1. **Pairwise Distances & Naive DBSCAN Workload:**
   MLSys_Worksheet2_Clas... +1
   1. Without spatial indexing, naive DBSCAN compares every point to every other point.
      MLSys_Worksheet2_Clas... +1
   2. Number of pairwise distance comparisons:

      MLSys_Worksheet2_Clas... +1

$$N_{\text{pairs}} \approx N^2$$

(Or $\frac{N(N-1)}{2}$ if symmetry is exploited, but standard matrix calculations in the hint use $\approx N^2$).
MLSys_Worksheet2_Clas... +3

3. Total FLOPs:

MLSys_Worksheet2_Clas... +1

$$\text{Total FLOPs} \approx N^2 \times (\text{FLOPs per distance})$$

MLSys_Worksheet2_Clas... +1

2. **Inference / Run Time Model:**

MLSys_Worksheet2_Clas... +1

$$\text{Run Time} = \frac{\text{Total FLOPs}}{\text{Effective Compute Throughput}} = \frac{\text{Total FLOPs}}{R_{\text{peak}} \times \eta}$$

MLSys_Worksheet2_Clas... +1

3. **Spatial Indexing Complexity:**
MLSys_Worksheet2_Clas... +1
    1. Using a spatial tree (e.g., k-d tree, Ball tree) or a spatial grid partitions the space so each point only checks points within its immediate local cells/branches.
    MLSys_Worksheet2_Clas... +1
    2. Time complexity drops from quadratic (N^2) down to $(N \log N)$.
    MLSys_Worksheet2_Clas... +1

Given Parameters (Question F2)

- Number of GPS points: $N = 2 \text{ million} = 2 \times 10^6$
MLSys_Worksheet2_Clas... +1
- FLOPs per distance calculation: $\approx 5 \text{ FLOPs}$
MLSys_Worksheet2_Clas... +1
- Peak compute rate: $R_{\text{peak}} = 1 \text{ TFLOP/s} = 10^{12} \text{ FLOP/s}$
MLSys_Worksheet2_Clas... +1
- Hardware efficiency factor: $\eta = 0.3$
MLSys_Worksheet2_Clas... +1

Step-by-Step Solutions (Question F2)

(a) Compute Total FLOPs, Run Time, and Spatial Index Effect

MLSys_Worksheet2_Clas... +1

1. **Calculate Total FLOPs:**
   MLSys_Worksheet2_Clas... +1
   1. Number of pairwise comparisons:

$$N^2 = (2 \times 10^6)^2 = 4 \times 10^{12} \text{ distance pairs}$$

      MLSys_Worksheet2_Clas... +1
   2. Total FLOPs:

$$\text{Total FLOPs} = 4 \times 10^{12} \times 5 \text{ FLOPs} = \quad \times \qquad \text{FLOPs}$$

      MLSys_Worksheet2_Clas... +1
2. **Calculate Run Time:**
   MLSys_Worksheet2_Clas... +1
   1. Effective compute throughput:

$$R_{\text{peak}} \times \eta = 1 \text{ TFLOP/s} \times 0.3 = 0.3 \text{ TFLOP/s} = 0.3 \times 1$$

      MLSys_Worksheet2_Clas... +1
   2. Run time:

$$\text{Run Time} = \frac{2 \times 10^{13} \text{ FLOPs}}{0.3 \times 10^{12} \text{ FLOP/s}} = \frac{20}{0.3} \text{ s} \approx \qquad \text{second}$$

      MLSys_Worksheet2_Clas... +1
3. **How a Spatial Index (k-d tree or grid) Changes the Scaling:**
   MLSys_Worksheet2_Clas... +1
   1. Instead of exhaustively testing all N^2 pairs, a spatial index prunes distant points and queries only points within the local ϵ-neighborhood.
      MLSys_Worksheet2_Clas... +1
   2. This reduces the algorithmic scaling from (N^2) down to $(N \log N)$.
      MLSys_Worksheet2_Clas... +1
   3. For $N = 2$ million, $N \log_2 N \approx 2 \times 10^6 \times 21 \approx 4.2 \times 10^7$ operations, making the query thousands of times faster (sub-second runtime).

MLSys_Worksheet2_Clas... + 1

PDF

MLSys Project Specifications

What good project ideas can be for this

Here are four strong project ideas designed specifically around the course rubric: spanning **two distinct tiers (Edge/Mobile/TinyML ↔ Server/Cloud)**, integrating real hardware available in the lab, and focusing on **systems architecture, data movement, and live demonstration reliability** rather than complex model engineering.

MLSys Project Specificati... + 3

1. Smart Asset Vibration Monitoring & Predictive Maintenance (TinyML ↔ Server)

- **Real-World Problem:** Industrial equipment (motors, pumps, conveyor rollers) suffers catastrophic mechanical failure due to bearing wear or imbalance. Continuously sending high-frequency 3-axis accelerometer data to the cloud wastes immense bandwidth.
- **Architecture & Flow:**
  - **Tier 1 (TinyML / Edge):** An Arduino Nano 33 BLE Sense or ESP32 with an IMU (borrowed from CTLab). It collects 3-axis accelerometer/gyro readings at 50–100 Hz. It runs an embedded **TensorFlow Lite for Microcontrollers (TFLite Micro)** model (a 1D-CNN or Autoencoder) that detects normal vs. anomalous vibration signatures locally.
    MLSys Project Specificati... + 2
  - **Trigger/Upload:** When an anomaly is detected, or on an hourly aggregate schedule, the MCU transmits sensor logs and anomaly telemetry via Wi-Fi/MQTT or serial/HTTP.
    MLSys Project Specificati...
  - **Tier 2 (Edge/Cloud Server):** A local laptop or cloud server ingesting anomaly logs. It runs automated batch re-clustering/re-training (e.g., Random Forest or Isolation Forest) to diagnose specific failure classes (bearing wear vs. rotor imbalance), visualizes equipment health on a dashboard, and tracks model drift.
    MLSys Project Specificati... + 1

- **Why it shines for grading:** It is easy to simulate live (shaking or tapping the board triggers the anomaly detection in Tier 1 and immediate alert streaming to Tier 2).
  MLSys Project Specificati...

2. Edge-Filtered Occupancy & Dynamic Energy Optimization (Edge Vision $\leftrightarrow$ Cloud)

- **Real-World Problem:** HVAC and smart lighting systems need real-time occupancy counts, but uploading raw 1080p video streams from every room introduces massive network overhead and major user privacy violations.
- **Architecture & Flow:**
  - **Tier 1 (Edge Device):** A Raspberry Pi (or Jetson Nano / edge laptop) equipped with a USB/Pi camera. It executes a lightweight, quantized detector (e.g., **YOLOv8-Nano** via ONNX Runtime or TFLite) to count visible humans in real time.
    MLSys Project Specificati... + 2
  - **Data Reduction:** Rather than streaming raw frames upstream, the edge node strips video and uploads only clean lightweight JSON payloads: `{"timestamp": ..., "room_id": "L201", "person_count": 3}`.
    MLSys Project Specificati...
  - **Tier 2 (Cloud / Central Server):** Ingests the time-series count data into a database (PostgreSQL/TimescaleDB). A cloud service trains/updates a time-series forecasting model (e.g., Ridge Regression, ARIMA, or a simple MLP) to forecast peak room occupancy hours and automate thermostat/lighting scheduling.
    MLSys Project Specificati... + 1
- **Why it shines for grading:** Provides a textbook architectural justification for your ACM report (privacy preservation + 99% bandwidth reduction).
  MLSys Project Specificati... + 1

3. Smart Recycling Sorter with Continual Learning Loop (Edge Camera $\leftrightarrow$ Cloud Retraining)

- **Real-World Problem:** Automated smart trash bins must classify waste (e.g., plastic bottle, paper cup, aluminum can, hazardous e-waste) instantly without relying on continuous internet access.
  MLSys Project Specificati...
- **Architecture & Flow:**
  - **Tier 1 (Edge / Mobile Tier):** A Raspberry Pi with camera or an Android/iOS phone (Mobile Tier) running a quantized MobileNetV2 / TFLite classifier.
    MLSys Project Specificati... + 1
  - **Local Inference & Active Learning Filter:** The local device classifies waste in $< 150\ \text{ms}$ to trigger bin solenoids. If prediction confidence is low (< 0.65), the device flags the item as an edge case, saves the compressed frame locally, and periodically syncs unconfident images to the cloud.
    MLSys Project Specificati... + 1

    - **Tier 2 (Cloud Server):** Receives unconfident/mislabeled samples. An automated pipeline stages new data, kicks off fine-tuning runs (PyTorch), evaluates validation metrics against the current baseline, and exports a new `.tflite` model to a model store for versioned over-the-air deployment.
      MLSys Project Specificati... + 2
- **Why it shines for grading:** Directly fulfills the prompt's requirement where Tier 2 explicitly **"Trains a more complex machine learning model based on the collected data and stores it for deployment"**.
  MLSys Project Specificati...

4. Acoustic Stress & Audio Anomaly Detection (Mobile App ↔ Cloud Backend)

- **Real-World Problem:** Detecting acoustic distress, glass breaks, baby crying, or machine squeals on resource-constrained mobile hardware without sending continuous raw audio feeds over cellular data.
- **Architecture & Flow:**
    - **Tier 1 (Mobile / Edge):** An Android app (or laptop/Pi audio node) continuously buffering audio through a ring buffer. It runs an on-device TFLite model converting audio into Mel-spectrograms and classifying ambient sound events locally.
      MLSys Project Specificati... + 1
    - **Telemetry:** Only when specific targeted sounds or high decibel anomalies occur does it stream the acoustic event metadata and short audio clips upstream via REST API.
      MLSys Project Specificati...
    - **Tier 2 (Cloud):** A FastAPI/Flask backend hosted on GCP or AWS EC2. It ingests raw audio clips, extracts higher-dimensional audio embeddings, and trains a broader multi-class classifier or autoencoder to discover new environmental acoustic clusters.
      MLSys Project Specificati... + 2
- **Why it shines for grading:** High portability during your 15-minute live demo—you can make an audio sound or play a YouTube clip from your phone to immediately demonstrate end-to-end data transfer to your backend.
  MLSys Project Specificati...

Recommended Tool Stack Matrix

Tier	Recommended Tools & Frameworks	Hardware Options
Tier 1 (Edge / Mobile / TinyML) MLSys Project Specificati...	TensorFlow Lite, ONNX Runtime Mobile, OpenCV, MQTT / HTTP clients MLSys Project Specificati... + 1	Raspberry Pi 4, ESP32, Jetson Nano, Android Phone (from CTLab) MLSys Project Specificati...

Tier	Recommended Tools & Frameworks	Hardware Options
Tier 2 (Cloud / Server) MLSys Project Specificati...	PyTorch, Scikit-Learn, FastAPI, SQLite / PostgreSQL, Docker MLSys Project Specificati... +1	Cloud VM (AWS/GCP), Docker container on local server MLSys Project Specificati...
Inter-tier Communication MLSys Project Specificati...	REST API (HTTP POST) or MQTT broker (Mosquitto) for lightweight sensor telemetry MLSys Project Specificati...	Local Wi-Fi or Ngrok tunnel MLSys Project Specificati...

Which domain or hardware setup (Embedded Microcontrollers, Raspberry Pi / Jetson vision, or Mobile) best matches the equipment you want to use?

Can I do something like detecting if a website is phising or not

Yes, absolutely. A phishing website detection system fits the **CEMCLRN / MAC-LRN multi-tier machine learning requirements.**

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To satisfy the rubric, you must ensure it is not just a standalone notebook, but an **end-to-end system spanning at least two distinct computing tiers**:

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1. **Tier 1 (Mobile / Edge / Browser Extension / Local Client):** Local lightweight inference and data capture.
   MLSys Project Specificati...
2. **Tier 2 (Cloud / Edge Server):** Ingestion, heavy feature extraction/retraining, and model management.
   MLSys Project Specificati...

Here is an architectural breakdown of how to make this project align with the course specifications:

System Architecture & Workflow

Tier 1: Client-Side Lightweight Filter (Mobile App, Edge Gateway, or Browser Extension)

- **Environment:** Mobile (React Native / Android with TFLite / ONNX Runtime Mobile) or a Chromium-based Browser Extension / Local Proxy Daemon.
- **Role (Fast Path / Local Inference):**
  - When a user visits or taps a link, Tier 1 intercepts the URL string and lightweight metadata (e.g., URL length, domain entropy, presence of IP address, suspicious TLD, subdomains count, HTTPS token misuse).
  - Runs a **quantized, lightweight pre-trained model** (e.g., a small Decision Tree / Random Forest converted via `m2cgen` into C/JS, or a quantized 1D-CNN via ONNX/TFLite).
    MLSys Project Specificati...
  - **Latency target:** < 10 ms. If it's an obvious benign URL or an obvious known phishing pattern, it provides an immediate verdict locally.
  - **Edge Case Telemetry:** If prediction uncertainty is high (e.g., probability score is ambiguous between 0.40 and 0.70) or the user manually reports an unrecognized suspicious site, Tier 1 packages the URL and metadata into a JSON payload and streams it to Tier 2 via REST API.
    MLSys Project Specificati...

Tier 2: Cloud / Central Server (Deep Analysis & Model Maintenance)

MLSys Project Specificati... +1

- **Environment:** Cloud VM (AWS EC2 / GCP Compute Engine / local Docker server) running a FastAPI / Flask backend.
  MLSys Project Specificati... +1
- **Role (Deep Path & Retraining Loop):**
  - **Data Ingestion:** Ingests flagged and ambiguous URLs sent from Tier 1 into a persistent store (e.g., PostgreSQL or SQLite).
    MLSys Project Specificati...
  - **Heavy Feature Extraction:** Performs deep server-side checks that are too resource-heavy for Tier 1:
    - Live WHOIS queries (domain registration age).
    - SSL certificate issuer and validity checks.
    - Scraping and parsing full HTML DOM structure, external script origins, and hidden form actions.
  - **Model Retraining:** Periodically retrains a heavier model (e.g., XGBoost, CatBoost, or a RoBERTa-based URL/HTML embedding model) using the aggregated edge data .
    MLSys Project Specificati...
  - **Model Distribution:** Distributes distilled, pruned, or newly calibrated lightweight weights back down to Tier 1 clients (or provides a cloud fallback prediction

endpoint).
MLSys Project Specificati...

Why This Fits the Course Requirements

1. **Multi-Tier Separation:** Clearly splits latency-critical lightweight edge inference from heavy server-side feature extraction and model retraining.
MLSys Project Specificati...
2. **Clear Justification for the ACM Report:**
    1. **Privacy & Bandwidth:** You don't stream every single website a user browses to the cloud (which would be a massive privacy risk and waste bandwidth). You only offload ambiguous cases.
MLSys Project Specificati... +1
    2. **Latency:** Users cannot wait 1–2 seconds for a cloud round-trip on every navigation click; sub-millisecond local filtering is required for UX.
3. **Hardware / Environment Requirements:** Satisfies the **Mobile** ↔ **Cloud** or **Edge** ↔ **Cloud** requirement cleanly.
MLSys Project Specificati...
4. **Demonstrable in 15 Minutes:**
    1. During the live presentation, you can feed a known benign link (e.g., `google.com`) and a synthetic phishing link (e.g., `secure-paypal-verify-login.xyz`) into your client.
MLSys Project Specificati...
    2. The audience sees instant local classification on Tier 1.
MLSys Project Specificati...
    3. You can trigger an ambiguous link to show the payload arriving live at your Tier 2 FastAPI server dashboard, followed by an automated retraining run updating the model store.
MLSys Project Specificati... +1

Readily Available Datasets

You do not need to collect phishing data from scratch to get started:

- **PhishTank API:** Free, constantly updated feed of verified phishing URLs.
- **OpenPhish:** Live zero-day phishing feeds.
- **Tranco / Majestic Million:** Verified top benign domains for negative training samples.
- **Kaggle Phishing Datasets:** Pre-extracted tabular datasets containing 30–50 lexical and structural URL features (e.g., *Malicious URLs Dataset*, *Phishing Websites Dataset*).

We have a subject called project proposal what should I put there?

Here is a clean, ready-to-use **Project Proposal** template and draft tailored directly to the **CEMCLRN / MAC-LRN Multi-Tier Machine Learning System** specifications.

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Whether "Project Proposal" is a section/subject in your document or an assignment submission on Canvas, you can structure it using the sections below:

Project Title

PhishGuard: A Multi-Tier Edge-to-Cloud Machine Learning System for Real-Time Phishing URL Detection and Continual Model Retraining

1. Problem Statement

Phishing attacks evolve rapidly, with malicious actors generating ephemeral domains that bypass static blocklists within hours. Conventional cloud-only URL scanners introduce high network latency ($500\ \text{ms}–2\ \text{s}$ per page load), consume excessive bandwidth, and compromise user privacy by transmitting a user's entire browsing history to a central server. Conversely, purely client-side models struggle to adapt to zero-day phishing campaigns without frequent updates and cannot perform heavy external network verifications (such as WHOIS registration lookups or full DOM scraping) on resource-constrained devices.

This project addresses this trade-off by designing a two-tier machine learning system that executes fast, privacy-preserving lexical URL inference on the client edge (Tier 1) while offloading uncertain edge cases and continuous model retraining to a centralized cloud server (Tier 2).

MLSys Project Specificati... +1

2. Proposed Two-Tier System Architecture

System Tier 1: Local Inference & Telemetry Collection (Mobile / Edge Client)

- **Environment:** Mobile application (or Edge/Browser Client) operating as a resource-constrained local gateway.

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- **Functionality:**
  - **Real-Time Local Feature Extraction:** Extracts lightweight lexical features directly from the requested URL string in sub-millisecond time (e.g., URL length, Shannon entropy, subdomain count, presence of raw IP addresses, suspicious TLDs, and `@` or `-` token anomalies).
  - **Lightweight Local Inference:** Runs a pre-trained, quantized lightweight model (e.g., a shallow Random Forest / Decision Tree or quantized 1D-CNN via **ONNX Runtime / TensorFlow Lite**) to classify links locally as *Benign* or *Phishing* in $< 10 \text{ ms}$.
    MLSys Project Specificati... +1
  - **Selective Data Collection:** When the local model outputs an ambiguous confidence score (e.g., $P(\text{phishing}) \in [0.35, 0.65]$) or a user flags a suspicious page, Tier 1 sends the URL and local feature vector to Tier 2 via a REST API.
    MLSys Project Specificati...

System Tier 2: Deep Analysis, Ingestion & Model Retraining (Cloud Server)

- **Environment:** Cloud/Edge Server hosting a **FastAPI** backend and **PostgreSQL / SQLite** database (containerized via Docker).
  MLSys Project Specificati...
- **Functionality:**
  - **Data Ingestion:** Collects flagged and ambiguous URL telemetry from Tier 1 devices alongside live threat feeds (e.g., PhishTank / OpenPhish).
    MLSys Project Specificati...
  - **Heavy Feature Enrichment:** Performs compute- and network-intensive extractions unsuitable for edge devices, including SSL certificate validation, DNS/WHOIS domain age lookups, and HTML DOM inspection.
  - **Model Training & Management:** Trains a higher-capacity classifier (e.g., **XGBoost** or **Scikit-Learn / PyTorch** ensemble) on the enriched dataset, versions the model artifacts, and compiles an updated lightweight model for over-the-air (OTA) deployment back to Tier 1 clients.
    MLSys Project Specificati... +1

3. Justification of Design Choices

- **Minimizing Data Transfer & Preserving Privacy:** Roughly $85\%–90\%$ of everyday web traffic goes to well-known benign domains. Filtering these locally on Tier 1 reduces upstream bandwidth consumption and protects user privacy by only transmitting uncertain or suspicious URLs to the cloud.
  MLSys Project Specificati...
- **Low-Latency User Experience:** Local inference avoids network round-trip delays, blocking malicious links immediately before the browser renders the page.
  MLSys Project Specificati...

- **Adaptability to Concept Drift:** Phishing patterns shift constantly. Tier 2 ensures the system is not static by continuously ingesting novel edge cases and retraining the model.
  MLSys Project Specificati...

4. Technology Stack & Datasets

- **Tier 1 (Edge/Mobile):** Python / JavaScript / Mobile client, TensorFlow Lite or ONNX Runtime.
  MLSys Project Specificati...
- **Tier 2 (Cloud Server):** FastAPI, Scikit-Learn / XGBoost / PyTorch, PostgreSQL/SQLite, Docker.
  MLSys Project Specificati...
- **Version Control:** Git / GitHub with structured branch workflows.
  MLSys Project Specificati...
- **Datasets:** Kaggle Malicious URLs Dataset, PhishTank API (verified phishing URLs), and Tranco Top Domains list (benign baseline).
  MLSys Project Specificati...

5. Evaluation Metrics

1. **Model Performance:** Precision, Recall, F1-Score, and False Positive Rate (critical to avoid blocking legitimate websites).
   MLSys Project Specificati...
2. **Systems Performance:**
   1. Tier 1 local inference latency (ms) and model memory footprint (KB/MB).
      MLSys Project Specificati...
   2. Bandwidth reduction ratio (% of requests resolved locally vs. offloaded to Tier 2).
      MLSys Project Specificati...
   3. End-to-end retraining and deployment pipeline execution time.
      MLSys Project Specificati... +1

Quick Tip if Submitting as a Short Form / 1-Paragraph Summary:

If the "Project Proposal" is just a short text box or email subject submission to Dr. Laguna, you can condense it to this:

> **Proposed Project: PhishGuard — Multi-Tier Phishing Detection System**
>
> We propose an end-to-end Edge-to-Cloud phishing detection system. **Tier 1 (Mobile/Edge Client)** extracts fast lexical URL features locally and runs a lightweight quantized model (TFLite/ONNX) for real-time ($< 10\ \mathrm{ms}$) link classification, preserving user privacy and minimizing network traffic. Ambiguous predictions and newly reported links are forwarded to **Tier 2 (Cloud Server)**,

which enriches URLs with heavy network/DOM features, logs data in a database, retrains a high-capacity model (XGBoost/PyTorch), and manages updated model weights for re-deployment to Tier 1.

MLSys Project Specificati... + 4

MLSys Project Specificati... + 4

can you make everything shorter

Project Title: PhishGuard: Multi-Tier Phishing URL Detection System

1. Problem Statement
Cloud-only phishing scanners add network latency and expose user browsing history, while static client-side models struggle to detect newly created zero-day phishing domains. This project builds a two-tier system that runs fast, privacy-preserving URL checks locally and offloads uncertain links to a cloud server for deep analysis and continual retraining.

MLSys Project Specificati... + 1

2. System Architecture

MLSys Project Specificati... + 1

- **Tier 1 (Mobile / Edge Client):** Extracts lightweight lexical features (URL length, subdomain count, entropy) and runs a quantized model (TFLite / ONNX) for instant classification (< 10 ms). Uncertain predictions or user-reported links are sent to Tier 2 .
  MLSys Project Specificati... + 2
- **Tier 2 (Cloud Server):** A FastAPI and PostgreSQL backend ingests flagged URLs, extracts heavy network/DOM features (WHOIS age, SSL certs), retrains a stronger classifier (XGBoost / PyTorch), and stores updated weights for deployment back to Tier 1.
  MLSys Project Specificati... + 3

3. Design Justification

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- **Privacy & Bandwidth:** Resolves $\approx 90\%$ of benign traffic locally, minimizing data transfer costs and keeping browsing history private.
  MLSys Project Specificati...
- **Low Latency:** Blocks known malicious links before the page loads without waiting for a cloud round-trip.

- **Continual Learning:** Tier 2 adapts to new phishing trends by retraining on edge-collected anomalies.
  MLSys Project Specificati...

4. Tech Stack & Datasets

MLSys Project Specificati... + 1

- **Stack:** Mobile/Browser Client, ONNX/TFLite (Tier 1); FastAPI, Scikit-Learn/PyTorch, PostgreSQL, Docker, Git (Tier 2).
  MLSys Project Specificati... + 3
- **Data:** PhishTank API, Kaggle Malicious URLs dataset, and Tranco Top Benign Domains.
  MLSys Project Specificati...

5. Evaluation Metrics

MLSys Project Specificati...

- **Model:** Precision, Recall, F1-Score, and False Positive Rate.
  MLSys Project Specificati...
- **System:** Local inference latency (ms), model size (KB), and percentage of requests resolved locally vs. offloaded.
  MLSys Project Specificati...

i am now planning on starting on this project. can you dissect the project into phases so i can allot time for them each day.

Here is a structured, 5-phase breakdown designed so you can chip away at one focused task per day and hit every requirement in the syllabus.

MLSys Project Specificati... + 1

Project Roadmap & Daily Schedule

Phase	Days	Focus Area	Key Deliver
Phase 1	Days 1–2	Repo Setup & Baseline Data	Public GitHu dataset, lexic extractor MLSys Projec
Phase 2	Days 3–5	Tier 1: Local Edge/Client	Quantized O model runnir

Phase	Days	Focus Area	Key Deliver
			locally MLSys Projec
Phase 3	Days 6–8	Tier 2: Cloud Server & Retraining	FastAPI back and automat script MLSys Projec
Phase 4	Days 9–10	End-to-End Integration & Benchmarks	Connected T pipeline + lat metrics MLSys Projec
Phase 5	Days 11–13	ACM Report & Live Demo Prep	5–7 page AC 15-minute liv flow MLSys Projec

Detailed Daily Breakdown

Phase 1: Setup, Data & Feature Engineering (Days 1–2)

- **Day 1: Git Setup & Dataset Curation**
  - Initialize your public GitHub repository with a clean folder structure (`/tier1_edge`, `/tier2_cloud`, `/data_pipeline`, `/models`) and a proper `.gitignore` to exclude raw CSVs and virtual environments.
    MLSys Project Specificati...
  - Download a baseline dataset (e.g., Kaggle Malicious URLs + Tranco top benign domains) so you have labeled URLs ready.
    MLSys Project Specificati...
- **Day 2: Lexical Feature Extraction Script**
  - Write a fast, dependency-light script that extracts ~10–15 string-level features from any URL (length, dot count, subdomain count, digit ratio, Shannon entropy, IP-in-URL flag, suspicious TLD flag).
  - Keep this script pure and modular—both Tier 1 and Tier 2 will use it.
    MLSys Project Specificati...

Phase 2: Tier 1 — Resource-Constrained Local Client (Days 3–5)

- **Day 3: Train & Export the Lightweight Model**
  - Train a fast initial model (e.g., Decision Tree, shallow Random Forest, or Logistic Regression/1D-CNN) on your lexical features.
    MLSys Project Specificati... +1

    - Convert and export the trained model into a portable format like **ONNX** (`.onnx`) or **TensorFlow Lite** (`.tflite`).
      MLSys Project Specificati... +1
- **Day 4: Build the Tier 1 Client Interface**
    - Build a lightweight client (a clean Next.js/React mobile-view web app, browser extension, or Python edge script) that takes a URL, extracts the lexical features, and runs local inference.
      MLSys Project Specificati...
- **Day 5: Confidence Thresholding & Escalation Logic**
    - Program the routing rule: if $P(\text{phishing})$ is between 0.35 and 0.65 (uncertain) or if the user clicks "Report Suspicious Site," package the URL and features into a JSON payload ready to send to Tier 2.
      MLSys Project Specificati...

Phase 3: Tier 2 — Cloud Server & Retraining Pipeline (Days 6–8)

- **Day 6: FastAPI Ingestion & Database Setup**
    - Set up your Tier 2 server (FastAPI + SQLite/PostgreSQL/Supabase).
      MLSys Project Specificati...
    - Create a `POST /api/v1/ingest` endpoint that receives flagged URLs from Tier 1 and logs them into the database.
      MLSys Project Specificati...
- **Day 7: Heavy Server-Side Feature Enrichment**
    - Add a server-side check on ingested URLs that would be too slow for Tier 1 (e.g., checking SSL certificate validity, basic HTTP status/redirect count, or WHOIS domain age).
- **Day 8: Model Retraining & Model Registry**
    - Build a `POST /api/v1/retrain` endpoint (or background task) that pulls the newly ingested records from the database, retrains a stronger model (e.g., XGBoost or PyTorch MLP), versions the new model artifact (`model_v2.onnx`), and exposes a `GET /api/v1/model/latest` endpoint so Tier 1 can pull updates.
      MLSys Project Specificati... +1

Phase 4: Integration, Testing & Benchmarking (Days 9–10)

- **Day 9: End-to-End Wiring & Dockerization**
    - Connect Tier 1 to Tier 2 over HTTP. Test the full loop: input a tricky URL on Tier 1 $\rightarrow$ watch it escalate to Tier 2 $\rightarrow$ trigger retraining $\rightarrow$ Tier 1 downloads the updated model.
      MLSys Project Specificati... +2
    - Ensure every team member has distinct, well-labeled commits in the Git history.
      MLSys Project Specificati...
- **Day 10: Collect System & ML Metrics (Crucial for Rubric)**

- Measure and log your numbers for the report:
    1. **Model metrics:** Accuracy, Precision, Recall, F1-score.
    MLSys Project Specificati...
    2. **System metrics:** Tier 1 inference latency (ms) vs. Tier 2 round-trip latency (ms), model file size (KB), and percentage of traffic resolved locally without hitting the cloud.
    MLSys Project Specificati...

Phase 5: Documentation & Live Demo Prep (Days 11–13)

- **Day 11: Architecture Diagram & ACM Report (Part 1)**
    - Draw a clean system architecture diagram showing Tier 1, Tier 2, data flow arrows, and model update paths.
    MLSys Project Specificati...
    - Draft the Problem Statement, System Design, and Design Justifications in the two-column ACM conference template.
    MLSys Project Specificati...
- **Day 12: ACM Report (Part 2 — Results & Challenges)**
    - Insert your latency tables, accuracy charts, and bandwidth savings calculations into the report (target 5–7 pages total).
    MLSys Project Specificati...
- **Day 13: 15-Minute Live Demo Rehearsal**
    - Prepare 3 ready-to-paste test URLs for your live presentation:
    MLSys Project Specificati...
    1. A clearly safe URL (resolved instantly on Tier 1).
    MLSys Project Specificati... +1
    2. A blatantly malicious URL (blocked instantly on Tier 1).
    MLSys Project Specificati... +1
    3. An ambiguous zero-day URL (triggers live escalation to Tier 2, shows up on your server dashboard, and triggers a model update).
    MLSys Project Specificati... +1

How many days or weeks do you have until the final submission deadline, and are you working solo or dividing this across teammates?